

TUSHAR PARIHAST

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PROFILE

Final-year B.Tech student in Artificial Intelligence aiming for an AI Engineer role. I build LLM-, NLP- and computer-vision-driven applications end to end: from model training and evaluation (XLNet with Hugging Face and PyTorch, TensorFlow/OpenCV vision models) to retrieval systems (TF-IDF, cosine similarity, inverted indexes) and production-style services (FastAPI, PostgreSQL, Supabase, n8n, Gemini API). Knowledgeable in RAG concepts, prompt engineering and information retrieval, with a focus on reliable, maintainable AI systems.

EDUCATION

Bachelor of Technology in Artificial Intelligence – Kathmandu University, Nepal 2022 – 2026
Relevant coursework: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Database Systems, Software Engineering, Data Structures & Algorithms

Higher Secondary Education (+2 Science) – United Academy, Nepal 2020 – 2022

TECHNICAL SKILLS

LLM / NLP	Transformers, XLNet, Hugging Face, Gemini API, RAG concepts, Prompt Engineering, Information Retrieval, TF-IDF, spaCy, NLTK
ML / Deep Learning	PyTorch, TensorFlow, Scikit-Learn, OpenCV, Computer Vision, Model Training & Evaluation, Data Augmentation
Data Science	Pandas, NumPy, Matplotlib, PCA / t-SNE, Data Cleaning, Feature Engineering, Jupyter Notebook
Programming	Python, C, C++, JavaScript, TypeScript, SQL
Backend & Data	FastAPI, PostgreSQL, Supabase, MySQL, SQLite, REST API integration
Tools & Web	Git, GitHub, Docker, n8n (automation), VS Code, React, React Native, Next.js, Tailwind CSS, Vite

AI / LLM & NLP PROJECTS

AI-Powered Automatic Doctor Appointment System

Stack: Next.js, TypeScript, Tailwind CSS, Supabase, PostgreSQL, n8n, Gemini API

- Built a full-stack AI healthcare appointment platform with separate workflows for patients, doctors and administrators.
- Designed an NLP symptom-intake and classification flow that maps symptoms to medical specializations and automates doctor matching.
- Integrated the Gemini API with n8n to automate symptom processing, appointment workflows and patient notifications; also built an agentic variant on the Gemini REST API with SQLite memory.
- Implemented scheduling logic and PostgreSQL storage for real-time doctor availability, reducing manual coordination between patients and doctors.

NLP Preprocessing & Document Embeddings Application

Stack: Python, FastAPI, spaCy, NLTK, Scikit-Learn, TF-IDF

- Built a modular FastAPI service for tokenization, lemmatization, stemming, POS tagging, named entity recognition and unified NLP preprocessing.
- Implemented TF-IDF document embeddings with cosine-similarity retrieval to surface the most relevant documents for user queries.
- Added vector-space analysis and 2D projection (PCA / t-SNE) for exploring text-retrieval and embedding workflows.

Information Retrieval System

Stack: Python, NLP

- Implemented text preprocessing, inverted/positional indexing, Boolean retrieval, and index compression using gap, Variable-Byte and Golomb encoding.
- Built TF-IDF vector-space retrieval and cosine-similarity document ranking, developing practical understanding of search and retrieval systems.

Transformer-Based NLP Model: XLNet

Stack: Python, PyTorch, Hugging Face Transformers, XLNet

- Trained and evaluated an XLNet model for an NLP classification task, covering pretrained model configuration, tokenization, training, evaluation and inference.
- Built practical understanding of transformer architectures and of adapting pretrained models to downstream tasks.

COMPUTER VISION & APPLIED ML PROJECTS

Traffic Sign Detection System

Stack: Python, OpenCV, TensorFlow, Deep Learning

- Built a deep learning pipeline for multi-class traffic sign recognition: preprocessing, resizing, augmentation, training, evaluation with classification metrics, and inference on unseen images.

Domestic Animal Image Classification

Stack: Python, TensorFlow, Computer Vision, Custom Dataset

- Built a multi-class image classifier on a custom dataset that I collected in the field and labeled by hand, capturing real-world variation in appearance, lighting and environment.
- Applied augmentation and preprocessing to improve robustness; contributed collected imagery to related computer vision projects.

Data Science: Exploratory Data Analysis

Stack: Python, Pandas, NumPy, Matplotlib, Jupyter Notebook

- Cleaned, filtered and analyzed a mobile-phone dataset (Pandas indexing, NumPy, Matplotlib) in reproducible Jupyter notebooks.

FULL-STACK & MOBILE PROJECTS

Heena Designs: Full-Stack Business & E-Commerce Platform

Stack: React, TypeScript, Tailwind CSS, Supabase, PostgreSQL

- Built and deployed a platform for a mehndi training and design studio: dynamic landing page, database-backed gallery, course listings and an e-commerce shop with a persistent guest cart and order-request workflow.
- Developed a Supabase/PostgreSQL admin dashboard for content, courses, products, gallery, orders, appointments, contact info and FAQs, with role-based access control, notifications and English / Simplified Chinese support.
- Designed a structured appointment and custom-design request workflow with responsive interfaces optimized for slower networks.

Vet Finder

Stack: React Native, Geoapify API

- Developed a location-aware mobile app for finding nearby veterinary clinics, integrating the Geoapify API for real-time location search and mapping.

ACHIEVEMENTS & ACTIVITIES

- 2026 – Selected Participant, eSewa Hackathon (National Level)
- 2025, 2026 – 2x Stall Lead, AI Convergence – National AI Meet
- 2025 – Participant, Hack for Nepal National Hackathon (National Level)
- 2025 – First Runner-Up, Aaviskar 2025 (IoT Hackathon)
- 2024 – Volunteer, World Youth Hackfest
- 2023 – Best Volunteer Award, AI Crusade

LANGUAGES

Nepali (Native) • Maithili (Native) • English (Professional) • Hindi (Conversational)